



Research article

Health indicator construction by quadratic function-based deep convolutional auto-encoder and its application into bearing RUL prediction[☆]

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ABSTRACT

As one of the most important components of machinery, once the bearing has a failure, serious catastrophe may happen. Hence, for avoiding the catastrophe, it is valuable to predict the remaining useful life (RUL) of bearing. Health indicators (HIs) construction plays a greatly important role in the data-driven RUL prediction. Unfortunately, most of the existing HIs construction methods need prior knowledge and few of them construct HIs from raw vibration signals. For dealing with the above issues, a novel quadratic function-based deep convolutional auto-encoder is developed in this work. The raw bearing vibration signals are first preprocessed by low-pass filtering. Then the cleaned vibration signals are input into the quadratic function-based DCAE neural networks for constructing HIs of bearings. Compared with AE, DNN, KPCA, ISOMAP, PCA and VAE, it is revealed that the proposed methodology can construct a better HI from the raw bearing vibration signal in terms of comprehensive performance. Several comparative experiments have been implemented, and the results indicate that the HI constructed by quadratic function-based DCAE neural network has stronger predictive power than the traditional data-driven HIs.

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1. Introduction

As one of the most crucial parts in rotary machinery, rolling bearings have been broadly applied to different kinds of mechanical devices in industrial field, such as high speed train, machine tool, aero-engine [1,2]. Because of the complicated operating condition and severe external environment, rolling bearings are prone to failures that would bring about machine halt, accuracy loss and even casualties [3,4]. In order to prevent the above mentioned problems, it is urgent and indispensable to predict the remaining useful life (RUL) of rolling bearing. With the predicted RUL, the maintenance schedule of related equipment can be reasonably determined, so as to reduce downtime and maintenance expenses, and avoid major safety accidents. Hence, the bearing RUL prediction is extremely significant to the rotating machinery [5,6].

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To estimate the rolling bearing RUL, model-based and data-driven methods were widely utilized to predict RUL [7,8]. The purpose of the model-based method is to establish a physical model for describing the degradation trend of the mechanical system or component, however, the establishment of the model is difficult for a complicated mechanical equipment under the various working condition, which may hinder the applicability of the model-based method [9]. Data-driven methods primarily adopt machine learning and deep learning technologies to uncover the mechanical system's dynamical characteristics with the measuring data [10]. A variety of neural networks have been successfully employed to estimate RUL. For instance, Qin et al. [11] put forward a new neural network based on the gated recurrent unit with two attention gates, named gated dual attention unit (GDAU), for predicting the bearing RUL. Chen et al. [12] presented a two-step solution to predict RUL of a nonlinear deterioration process. Liu et al. [13] proposed Bi-directional gated recurrent unit (BGRU) with feature-attention mechanism to estimate RUL of the turbofan engines. Compared with the model-based method, there is no expert knowledge in the data-driven method, thus it has broad development prospects in practical engineering.

As the critical step in the data-driven RUL prediction, the construction of a proper health indicator (HI) will influence the accuracy and stability of RUL estimation to a large extent. To

describe the degradation trend, several HI construction methods have been developed. Chen et al. [14] utilized the relative root mean square (RMS) to assess the bearing degradation and divided the whole bearing life into several stages. The remaining life of bearing was predicted using the multivariable support vector machine, and the results verified the accuracy of the proposed method. Lei et al. [15] presented an approach based on distance metric learning (DML) to construct HIs of bearings with multiple optimal features extracted from monitoring data, and this new HI construction method was validated by bearing accelerated degradation dataset. Zhang et al. [16] studied the selection of degradation features using correlation, monotonicity and robustness. The feature selection process was regarded as a combinational constrained issue to solve. The experimental results demonstrated that this feature selection was greatly effective. A feature-based greedy kernel principal components analysis (KPCA) approach was proposed to reveal the nonlinear degradation trend by Chen et al. [17], which took the first component as the degradation feature of complex equipment. This unsupervised method was much effective for identifying the deterioration level. Huang et al. [18] explored a reliable HI construction method for bearing degradation assessment. Firstly, the correlation kurtosis for different iteration periods was calculated from frequency-domain data, and then it was fused by a Gaussian process latency variable model (GPLVM). Finally, the sensitive features obtained through GPLVM were chose to form the HI vector, which was utilized for RUL prediction with a multiple convolutional long short-term memory (MCLSTM) network. Most of the above conventional techniques can be just applied to handle with the dataset with a distinct degradation trend, and the potential degradation characteristics might not be effectively mined.

Aiming at learning the potential information from the monitoring data, various deep neural networks have been generally used for HI construction, which could automatically construct HI without the prior knowledge [19]. Wu et al. [20] built a multi-scale convolutional neural network (MSCNN) to construct a single HI for describing the bearing degradation state with the frequency-domain data. The MSCNN model could construct HI without any previous work and expertise. In addition, the constructed HI was able to better represent the degradation trend of bearing compared with other three HI construction methods. Chen et al. [21] proposed the end-to-end convolution recurrent neural network (RNN) for HI construction, which combined the structural superiorities of CNN and RNN. The CNN network was used to extract locally sequential feature and the RNN network was employed for connecting these features via the time-series information. Using polynomial decaying learning rate, an improved deep convolutional neural network (PDCNN) was proposed by She et al. [22], and the original vibration signals were input into PDCNN for HI construction. For this method, the staged degradation was considered and the failure threshold was easy to set in the process of RUL prediction. Yang et al. [23] researched an efficient approach to monitor the degradation of rolling bearings and predict the bearing RUL. The bearing HI was constructed by the self-normalizing convolutional neural network and the gated recursive unit (SCNN-GRU) with the high-quality features, which were selected from the complete feature set via three indexes: time correlation, monotonicity and robustness. The experimental results indicated that the HI based on SCNN-GRU could better represent the bearing degradation, and the RUL prediction error was the lowest compared to other methods. Auto-encoder (AE) has the ability to learn effective data coding by an unsupervised learning manner [24,25]. Learning a useful representation from input data is the fundamental purpose of AE, especially for reducing the data dimension. Hence, AE was

successfully applied to construct HIs of mechanical equipment, gear and bearing [26,27]. The ensemble stacked auto-encoders were proposed to construct the bearing HI by Lin et al. [28]. Four stacked activation-diverse auto-encoders were used for extracting features from vibration frequency spectra, and then these features were selected as the input of a deep neural network, which served as a non-linear function to reduce the feature dimension to one-dimension. Yang et al. [29] put forward an approach to automatically extract bearing HI based on sparse auto-encoder (SAE). The obtained results showed that the satisfactory HIs were automatically generated by various signals, without requiring any expert knowledge. Hemmer et al. [30] developed a novel approach for computing the low-speed axial rolling bearing HIs by using conditional variational auto-encoders (CVAEs), which did not need expert knowledge and could construct efficient condition indicators. She et al. [31] utilized sparse auto-encoder with regularization network to compress the raw features for obtaining some new features, and these new features were sorted by the trendability, and the HI was constructed by those selected features with large trendability. Ping et al. [32] developed a variational auto-encoder network (VAE) based on logarithmic normal distribution to extract the final feature with the lognormal prior information by multi-source monitoring data. Several experiments indicated that taking account of the deterioration features have the substantial heteroscedasticity and adding regularization items with prior information is greatly significant to extract more effective degradation feature. Xu et al. [33] made use of the stacked auto-encoders to detect the initial degradation process by frequency spectra of bearings, then employed an exponential function to construct the monotonous HI. Moreover, the proposed method did not rely on any prior knowledge, and it was superior to the existing HI construction methods. Chen et al. [34] proposed a new data-driven HI construction method based on an improved non-dominated sorting genetic algorithm-II (NSGA-II) and stacked auto-encoder (SAE). The improved NSGA-II was applied to optimize the candidate features extracted from the monitoring data. Then SAE was used to construct the HI by the optimized features. The experimental results showed that this approach could exactly recognize different equipment statuses and effectively decrease the complexity of the diagnostic model.

From the above works, it can be easily noted that few of the HI construction approaches directly generate HIs from raw vibration signals and did not consider the global degradation trend of bearing HI that is crucial for the accuracy of RUL prediction. Furthermore, the overall performance of HI including monotonicity, correlation, and robustness need be further improved. Whereupon, a deep convolutional auto-encoder (DCAE) neural network is proposed for extracting HIs from raw vibration signals of bearings. Meanwhile, a quadratic function is developed for labeling the variation tendency of HIs from an overall point of view, and a new loss term is added into the loss of DCAE for improving the monotonicity of the degradation trend. With the HI constructed by quadratic function-based DCAE, gated attention unit (GAU) neural networks are employed to predict the RULs of bearings. The contributions of this work are listed below.

(1) A novel HI construction method based on auto-encoder neural network, named deep convolutional auto-encoder (DCAE), is explored. Unlike various traditional HI construction methods, it can automatically extract the HIs of the bearings from the raw vibration signals without expert knowledge.

(2) A quadratic function is designed for labeling the output of DCAE (i.e. HI), and the loss function of DCAE is improved by introducing a new loss term, which can guarantee the monotonically decreasing trend of bearing HI and improve its monotonicity.

(3) The failure threshold of bearings with general applicability can be easily determined with the proposed HI construction approach, and the experimental results of several bearing

datasets demonstrate its superiority over other HI construction approaches including AE, deep neural network (DNN), KPCA, isometric mapping (ISOMAP), principal components analysis (PCA) and VAE.

2. Preliminary

As a neural network, AE is first proposed in Ref. [35] to reconstruct the input by an unsupervised manner, which consists of two operations: encoding and decoding. The encoding operation is used for transforming the input data from high dimension to low dimension, while the decoding operation converts the data of hidden layer from low dimension to high dimension as the output of AE. The weight matrix in AE is updated by the classical back propagation algorithm. The network architecture of AE is displayed in Fig. 1.

Suppose that $\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]^T$ denotes the input data. The output of hidden layer $\mathbf{z}$ is calculated by

$$\mathbf{z} = s_1(\mathbf{w}_1\mathbf{x} + \mathbf{b}_1) \quad (1)$$

where s_1 denotes an activation function used in encoding, and $\mathbf{w}_1$, $\mathbf{b}_1$ are respectively the weight and bias matrix of adjacent neurons. On the other hand, the output of the decoder $\mathbf{x}' = [x'_1 \ x'_2 \ \dots \ x'_n]^T$ is obtained as

$$\mathbf{x}' = s_2(\mathbf{w}_2\mathbf{z} + \mathbf{b}_2) \quad (2)$$

where s_2 denotes an activation function used in decoding, and $\mathbf{w}_2$, $\mathbf{b}_2$ are the weight and bias matrix of adjacent neurons, respectively. The weight and bias matrices of AE are optimized by minimizing the following loss function, which is defined as

$$J = \frac{1}{n} \sum_{j=1}^n \|x_j - x'_j\|^2 \quad (3)$$

3. The proposed HI construction methodology

3.1. Design of deep convolutional auto-encoder (DCAE)

By introducing convolution operation into the basic AE, DCAE neural networks are designed for generating bearing HIs. The convolution and pooling operation is first implemented on the input data, named encoding. Then, the input data is reconstructed by upsampling and convolution, named decoding. The structure of DCAE designed in this study is illustrated in Fig. 2.

As depicted in Fig. 2, DCAE contains twenty-seven layers: an input layer, an output layer, thirteen convolutional layers C1 to C13, six pooling layers P1 to P6, and six upsampling layers U1 to U6. As the most essential part of the DCAE, each convolutional layer contains some kernels. In general, the length of a data series is much bigger than that of a kernel. It follows that a small piece of signal is processed by the kernel in the convolutional layer. Suppose that $\mathbf{H}_{p,q}$ denotes the p th block data at layer q , and N_q is the length of such block data. $\mathbf{H}_{p,q}$ is split into $\mathbf{H}_{p,q}^{i:i+L}$, where L is the length of split window, and i represents the serial number in the block data. Then the convolution operation in the DCAE neural network is given by

$$\mathbf{Y}_{k,q+1}^i = s \left(\sum_{p=1}^{N_q} \mathbf{H}_{p,q}^{i:i+L} * \mathbf{w}_{k,p,q} + \mathbf{b}_{p,q} \right) \quad (4)$$

where $*$ represents one-dimensional convolution. $\mathbf{w}_{k,p,q}$ denotes the weight matrix between the p th block data at the q th layer and the k th block data at the $q + 1$ layer. $\mathbf{b}_{p,q}$ denotes the bias

matrix. $\mathbf{Y}_{k,q+1}^i$ is the output of convolution operation, where s is a nonlinear activation function, such as ReLU or Sigmoid.

As for the pooling layer, the output of the convolutional layer is first split into some neighborhood region, and then the maximum value within each neighborhood region is calculated by the max pooling function, which is given by

$$p_{n,m}^i = \max \left(\mathbf{Y}_{n,m}^{i:L_p:(i+1)*L_p} \right) \quad (5)$$

where L_p is the length of the pooling region. $p_{n,m}^i$ denotes the pooling result of the i th point at the n th block data of the m th layer.

The upsampling layer is used for decoding. The data dimension in the decoding is expanded by

$$U_{n,m}^i = \text{upsampling} \left(p_{n,m}^i \right) \quad (6)$$

where $U_{n,m}^i$ denotes the input of the i th point at the n th block data of the m th layer. $U_{n,m}^i$ denotes the output of the upsampling layer of the i th point at the n th block data of the m th layer.

Suppose that $\mathbf{X} = [X_1 \ X_2 \ \dots \ X_n]^T$ and $\mathbf{X}' = [X'_1 \ X'_2 \ \dots \ X'_n]^T$ denote the input and output of DCAE neural network respectively. The loss function of DCAE is defined as

$$J = \frac{1}{n} \sum_{j=1}^n \|X_j - X'_j\|^2 \quad (7)$$

3.2. Procedure of HI construction

The health degree of bearing in the life cycle is constantly changing. A proper representation for the degradation trend is useful to monitor the health state of bearing and predict the remaining useful life of bearing. Therefore, it is essential for designing a labeling function to describe the health degree. The conventional methods used the linear function or piecewise function to simulate the degradation trend of bearing. However, the two functions have some drawbacks.

As shown in Fig. 3(a), the linear function is employed to simulate the linear degradation trend of bearing, which is defined as

$$f(t_i) = -\left(\frac{1}{t_n} * t_i\right) + 1 \quad (8)$$

where t_n is the full lifetime of bearing and t_i is the current time in the bearing life cycle. In Fig. 3(a), the trend of the degradation rate is linear, which illustrates the degradation rate is steady. However, the degradation rate of bearing will not remain unchanged during the operation of bearing.

The piecewise function is designed to describe the degradation process of bearing, which is illustrated in Fig. 3(b). Its definition is written as

$$f(t_i) = \begin{cases} 1 & t_i \leq t_j \\ \left(\frac{1}{t_j - t_n}\right) * t_i + \left(\frac{t_n}{t_n - t_j}\right) & t_i > t_j \end{cases} \quad (9)$$

where t_j is the initial degradation time of bearing, i.e. the initial failure point. From Fig. 3(b), the health degree always keeps the same before the time t_j , and then linearly descends until the time of t_n , i.e. the total failure of bearing. However, it is extremely difficult to detect the initial failure, so we cannot effectively determine the time t_j . As previously mentioned, the degradation rate is probably different under different health statuses. Therefore, it is unreasonable to use the linear function to reflect the degradation trend of bearing, and a function that can better represent the degradation trend need be designed.

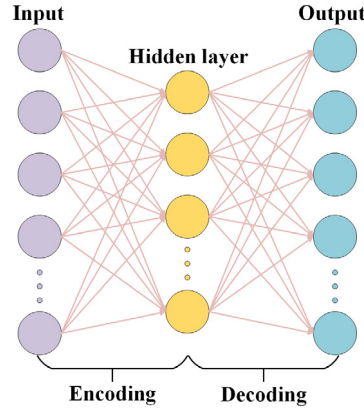


Fig. 1. The architecture of AE.

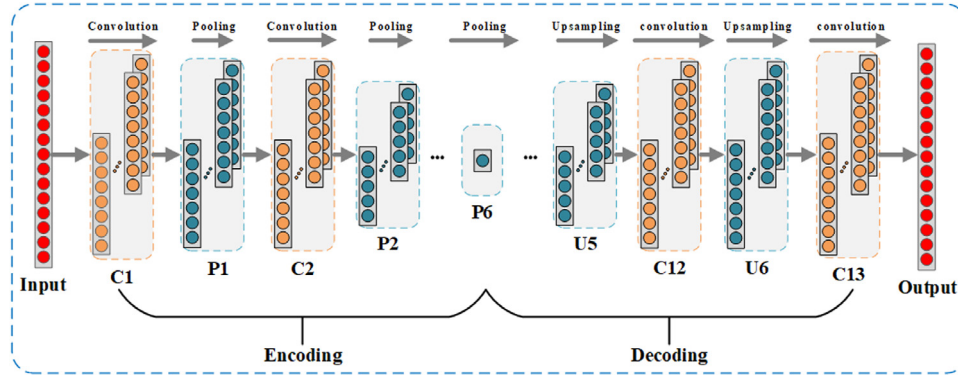


Fig. 2. The structure of DCAE.

Inspired by the law of human aging, this work considers that the degradation rate of the bearing's health should gradually increase with the increase of the run time. Hence, a quadratic function is designed to describe the degradation trend of bearing, which is defined as

$$f(t_i) = -\left(\frac{1}{t_n^2} * t_i^2\right) + 1 \quad (10)$$

The degradation trend described by Eq. (10) is shown in Fig. 3(c). From this figure, it can be seen that the degradation rate is small in the early stage due to the low failure probability of a new bearing. After the bearing runs a period of time under a certain load, several minor damages appear, and the degradation rate becomes larger due to the influence of damages. In the later stage of bearing life cycle, the major failure occurs, and it accelerates the deterioration of bearing's health. During this stage, the bearing has the fastest performance degradation speed. By comparing the three functions, the quadric function can better represent the degradation process of bearing's health. Moreover, the quadric function need not determine the inflection point. Via the proposed quadric function, a new loss can be added into the training of DCAE, so as to obtain the monotonically decreasing HI.

With the established DCAE and the proposed quadric function, HI can be automatically extracted. The raw vibration signal is fed into the encoder of DCAE to extract HI, and the output of encoder is used as the HI. To ensure the monotonically decreasing tendency of HI that accords with the actual degradation law, the proposed quadratic function is used to label the output of encoder. Fig. 4 illustrates the flowchart of the new HI construction methodology.

The concrete steps of the proposed HI construction methodology are summarized as follows.

(1) In the bearing accelerated degradation test, the raw vibration signals are collected by acceleration transducer. Let $\mathbf{x} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_N]^T$ represent a raw vibration signal dataset, where $\mathbf{x}_i = [x_i^1 \ x_i^2 \ \dots \ x_i^M]^T$, N denotes the length of vibration samples, and M denotes the number of each sample.

(2) The raw vibration dataset is preprocessed by low-pass filtering to suppress the influence of noise in the acquired vibration signal. Then the preprocessed vibration signal dataset $\mathbf{s} = [\mathbf{s}_1 \ \mathbf{s}_2 \ \dots \ \mathbf{s}_N]^T$ is obtained, where $\mathbf{s}_i = [s_i^1 \ s_i^2 \ \dots \ s_i^M]^T$.

(3) To obtain a decreasing degradation trend, the labels of encoding output (i.e. HI) $\mathbf{L} = [l_1 \ l_2 \ \dots \ l_N]^T$ in the training process is denoted by Eq. (8).

(4) By using the preprocessed vibration dataset $\mathbf{s}$ and HI labels $\mathbf{L}$, DCAE is trained. The output of the encoder $\mathbf{L}' = [l'_1 \ l'_2 \ \dots \ l'_N]^T$ and the output of DCAE $\mathbf{y} = [\mathbf{y}_1 \ \mathbf{y}_2 \ \dots \ \mathbf{y}_N]^T$ are expected to be as close as possible to the labels $\mathbf{L}$ and the input of DCAE $\mathbf{s}$ by minimizing the loss function

$$J = \frac{1}{N} \sum_{i=1}^N \|\mathbf{l}_i - \mathbf{l}'_i\|^2 + w * \frac{1}{N} \sum_{i=1}^N \|\mathbf{s}_i - \mathbf{y}_i\|^2 \quad (11)$$

where J is the loss function, w is proportionality coefficient. Using the training dataset, the weight and bias parameters in the DCAE neural network are updated via the minimization of Eq. (11). Via several epochs, the error curve of the DCAE neural network tends to be steady, which indicates that the training of DCAE is complete. Then, it will be used for constructing HIs of test bearings.

(5) The raw vibration signal dataset of the test bearing is also preprocessed by low-pass filtering. The preprocessed dataset is

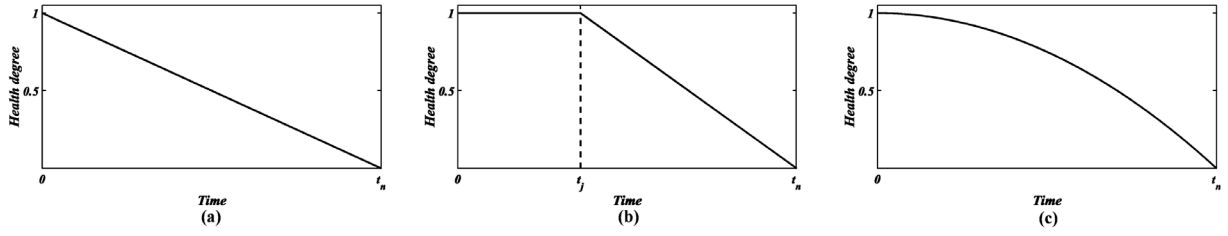


Fig. 3. Three function curve to describe the degradation trend of bearing.

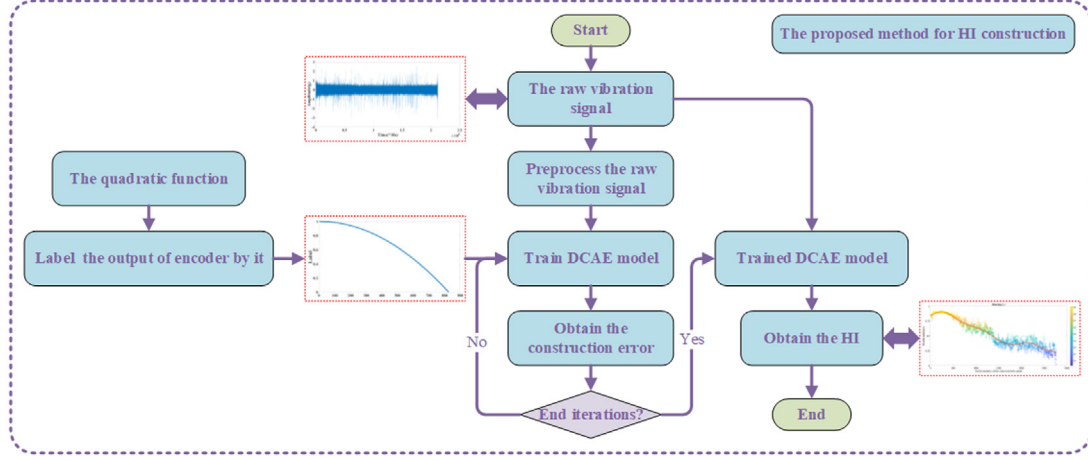


Fig. 4. The HI construction methodology proposed by this work.

$\mathbf{z} = [\mathbf{z}_1 \ \mathbf{z}_2 \ \dots \ \mathbf{z}_K]^T$, where $\mathbf{z}_i = [z_i^1 \ z_i^2 \ \dots \ z_i^M]^T$ and the encoder of the trained DCAE neural network is defined as f . Next, we feed the dataset $\mathbf{z}$ into the trained DCAE, and then obtain the output of encoder $\mathbf{h} = [h_1 \ h_2 \ \dots \ h_K]^T$, which is denoted by

$$\mathbf{h} = f(\mathbf{z}) \quad (12)$$

It immediately follows that the HI vector of the test bearing $\mathbf{h} = [h_1 \ h_2 \ \dots \ h_K]^T$ is calculated.

4. Comparison of different HI construction methods

4.1. Data description

The full lifecycle rolling bearing dataset from the PHM 2012 challenge is adopted to validate the proposed DCAE neural network. The dataset was obtained from a PRONOSTIA test rig, as shown in Fig. 5. By setting three working conditions, seventeen life-cycle tests of bearings were implemented, it then follows that seventeen life cycle datasets of raw vibration signals are acquired by accelerometers installed in both horizontal and vertical directions. The sampling frequency is 25.6 kHz, and a signal with the length of 0.1 s is collected every 10 s in the experiment. More details can be seen in Ref. [36]. The detailed experimental information and seventeen datasets are listed in Table 1. As listed in this table, the run-to-failure bearing1_1, bearing1_3, bearing2_3, bearing2_6 and bearing3_3 under three working conditions are chose to construct the test datasets, while the others are used to construct the training datasets.

4.2. Assessment index

To compare the DCAE method with other traditional methods, such as KPCA, DNN and AE, some appropriate assessment indexes are required. Thus monotonicity, correlation and robustness are

Table 1

The classification of seventeen bearing datasets under different working conditions.

Working condition	Condition1	Condition2	Condition3
Rotating speed (rpm)	1800	1650	1500
Loading force (N)	4000	4200	5000
Training dataset	Bearing1_2	Bearing2_1	Bearing3_1
	Bearing1_4	Bearing2_2	Bearing3_2
	Bearing1_5	Bearing2_4	
	Bearing1_6	Bearing2_5	
	Bearing1_7	Bearing2_7	
Testing dataset	Bearing1_1	Bearing2_3	Bearing3_3
	Bearing1_3	Bearing2_6	

employed to quantitatively assess the ability of one HI. For this goal, the polynomial fitting method is first used to decompose HI into an average trend and a random part:

$$H(t_n) = H_T(t_n) + H_R(t_n) \quad (13)$$

where $H(t_n)$ denotes the value of HI at time t_n , $H_T(t_n)$ denotes its mean trend and $H_R(t_n)$ denotes the random part.

Three HI assessment indexes including monotonicity (*Mon*), correlation (*Corr*) and robustness (*Rob*) are respectively formulated as

$$Mon(H) = \left| \frac{No. \text{ of } dH_T(t_n) > 0}{B-1} - \frac{No. \text{ of } dH_T(t_n) < 0}{B-1} \right| \quad (14)$$

$$Corr(H, T) = \frac{\left| \sum_{n=1}^N (H(t_n) - \bar{H})(t_n - \bar{T}) \right|}{\sqrt{\sum_{n=1}^N (H(t_n) - \bar{H})^2 \sum_{n=1}^N (t_n - \bar{T})^2}} \quad (15)$$

$$Rob(H) = \frac{1}{N} \sum_{n=1}^N \exp\left(-\left|\frac{H_R(t_n)}{H(t_n)}\right|\right) \quad (16)$$

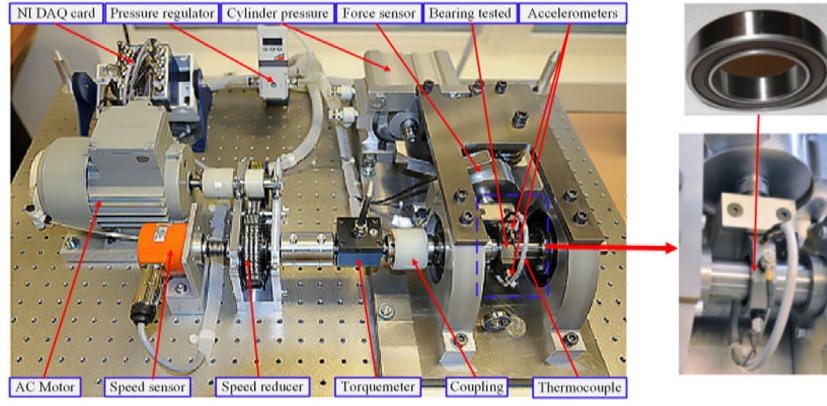


Fig. 5. PRONOSTIA test rig.

where $dH_T(t_n)$ is the derivative of $H_T(t_n)$, B denotes the length of the HI vector, $\bar{H} = (1/N \sum_{n=1}^N H(t_n))$, and $\bar{T} = (1/N \sum_{n=1}^N t_n)$.

Monotonicity index evaluates the variation tendency of a HI, including the increasing trend and the decreasing trend. The linear correlation between a HI vector and the corresponding time is measured by correlation. The tolerance of a HI to outliers is evaluated using robustness. Furthermore, it can be noted that all the three assessment indexes are confined in the range of [0,1]. To assess the overall ability of a HI, a composite index (CI) that contains all the three indexes is defined as

$$CI = 0.3 * Corr + 0.4 * Mon + 0.3 * Rob \quad (17)$$

4.3. Experiment results and comparison

The specific structure of the proposed DCAE is illustrated in Fig. 6. In the seven convolutional layers of the encoder, the numbers of convolution kernels are 20, 40, 80, 40, 20, 10 and 1 respectively. Alternatively, the numbers of convolution kernels are separately 10, 20, 40, 80, 40 and 20 in the seven layers of the decoder.

In the proposed DCAE, there are some parameters need to be determined, such as the proportionality coefficient w and the learning rate. Then, we will discuss how to select these parameters under different working conditions. To evaluate the capability of the trained DCAE neural network, a bearing dataset is chose from the test datasets, during the selection of parameters under different working conditions. First of all, the learning rate is regarded as a significant variable to search the proper learning rate of DCAE neural network. The experiment on each learning rate is repeated 10 times under three working conditions, and the composite indexes are computed. The boxplots of the results obtained by multiple learning rates are depicted in Fig. 7. We know from this figure that the optimal learning rate is 0.001 in both condition_1 and condition_2, while a better learning rate is 0.006 in condition_3. Similarly, the experiments for selecting the proportion coefficient w under different working conditions are implemented. The boxplots of the composite indexes obtained by multiple proportionality coefficients are shown in Fig. 8. We can know from this figure that the optimal proportionality coefficients w for three working conditions are 0.2, 1.0 and 0.4 respectively.

With the appropriate learning rate and the proportion coefficient, the construction of DCAE neural network is complete. After training DCAE neural network by the training datasets, the HIs of the test datasets are generated by the trained DCAE neural network. Fig. 9(a), (b), (c), (d) and (e) illustrate the HI curves of bearing1_1, bearing1_3, bearing2_3, bearing2_6 and bearing3_3,

respectively. Those curves respectively contain 2761, 2325, 1420, 687 and 425 HI points. It is easy to observe from all HI curves that the value of HI decreases as the serial number of characteristic point increases. Then we can conclude that the proposed HI based on quadratic function is able to effectively describe the degradation process of these bearings.

For further demonstrating the superiority of the proposed HI construction method, five HI construction methods including AE, DNN, KPCA, ISOMAP and PCA, are used for comparison. Four assessment indexes, *Mon*, *Corr*, *Rob*, and *CI*, are applied to quantitatively evaluate the three HIs. The evaluation results of *Mon*, *Corr* and *Rob* for all approaches are shown in Fig. 10, while the evaluation results of *CI* obtained by all methods are shown in Fig. 11. It can be noted from Fig. 10 that the indexes *Mon* and *Corr* of DCAE are greatly higher than those of AE, DNN, KPCA, ISOMAP and PCA, and its average *Rob* also exceeds those of other methods. In Fig. 11, we can see that the *CI*s obtained by the quadratic function-based DCAE are much higher than those obtained by other methods. From Figs. 10 and 11, we conclude that the DCAE neural network has the best performance among these approaches. Especially for *Mon*, *Corr* and *CI*, DCAE neural network is greatly superior to other methods. This is mainly because that the convolution and pooling operation is beneficial to extract potential information from high-dimensional data that is significant for the process of HI construction, and the quadratic function is used for labeling HI. Moreover, the HI values of all bearings are close to 0 once they lose efficacy. However, for other HI construction methods, the failure thresholds of different experiments often have a large variation. Thus it is more easy and convenient to determine the failure threshold of the rolling bearing by the proposed methodology. This is also a prominent advantage of the proposed HI. Therefore, the quadratic function-based DCAE neural network is more suitable for HI construction than AE, DNN, KPCA, ISOMAP and PCA.

As one of the extended forms of AE, VAE and its variants have been effectively applied to construct HIs. Therefore, it is indispensable to study the HI construction approach based on VAE neural network and quadratic function. With the vibration data of rolling bearings bearing1_1 and bearing1_3, the HI curves obtained by VAE are separately illustrated in Fig. 12(a) and (b). From Fig. 12, it is observed that the obtained HI curves are highly oscillatory and cannot represent the degradation trend, thus the quadratic function-based VAE neural network is unfit for constructing the bearing HI from the raw vibration signals.

5. Application into bearing RUL prediction

After obtaining the HI vector by the measured bearing dataset, an appropriate prediction model need be chose for RUL prediction. Essentially, the time series prediction network can predict

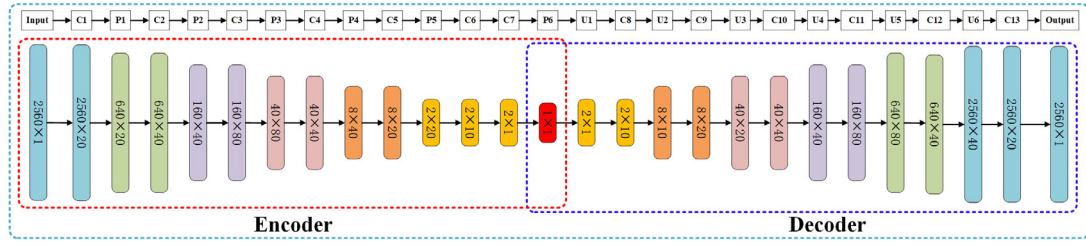


Fig. 6. The specific structure of DCAE.

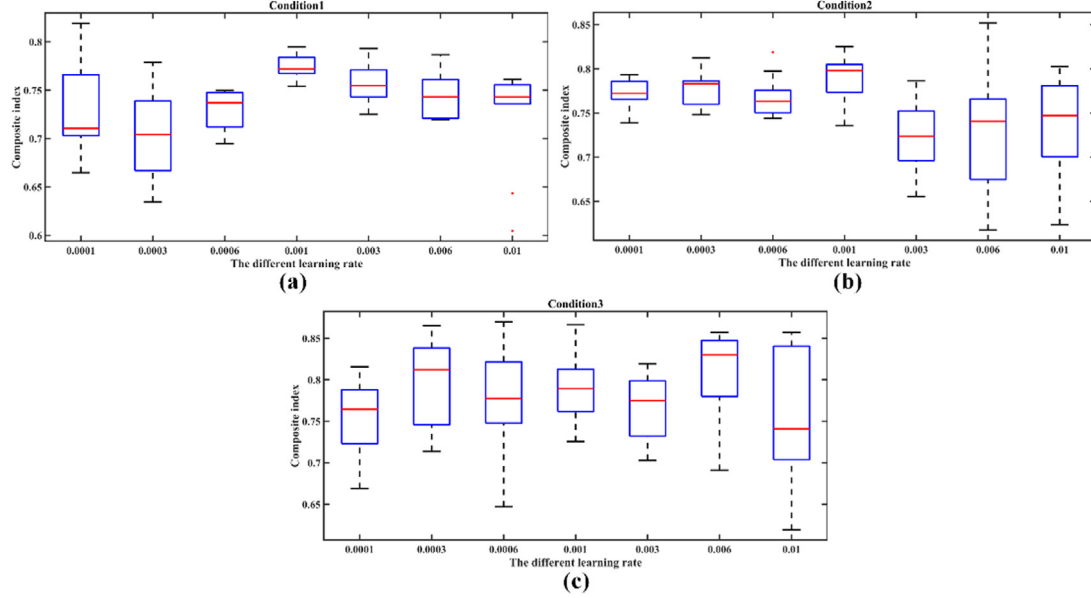


Fig. 7. Composite indexes obtained by multiple learning rates: (a) condition1, (b) condition2, (c) condition3.

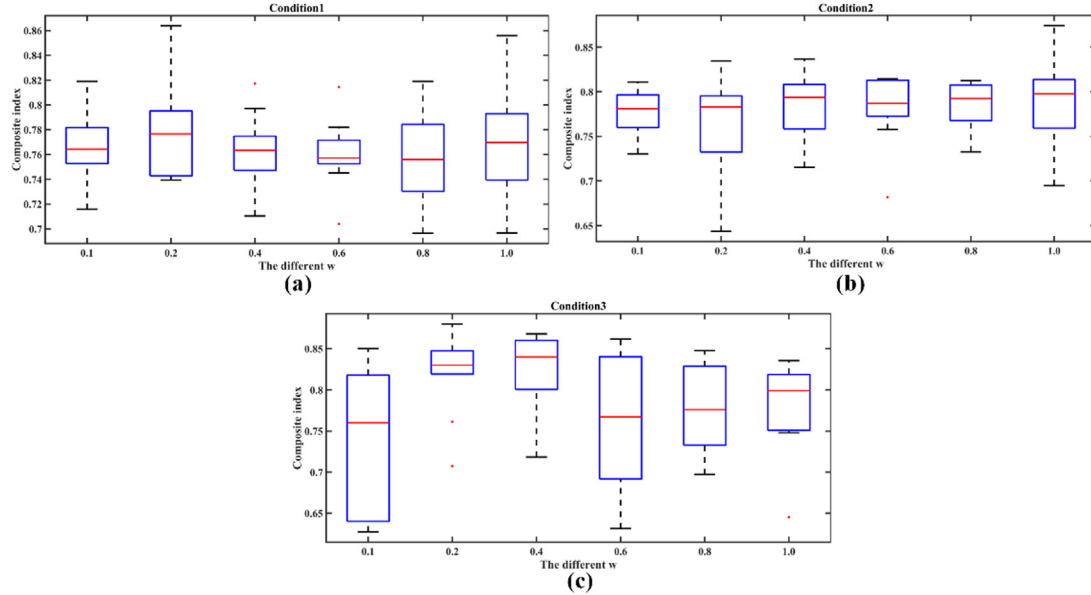


Fig. 8. Composite indexes obtained by multiple proportionality coefficients: (a) condition1, (b) condition2, (c) condition3.

the unknown HI values according to the known HI values that are obtained by the current (existing) data. In various time series prediction networks, recurrent neural networks (RNNs) have been widely used for the time series prediction. By introducing the gating mechanism, LSTM and its variants can properly remember

short-term and long-term information from monitoring data, so as to enhance the predictive ability of RNN. As the most popular variant of LSTM, GRU and its variants can effectively avoid the problem of vanishing gradients and exploding gradients. In particular, a recent predictive neural network, gated attention unit

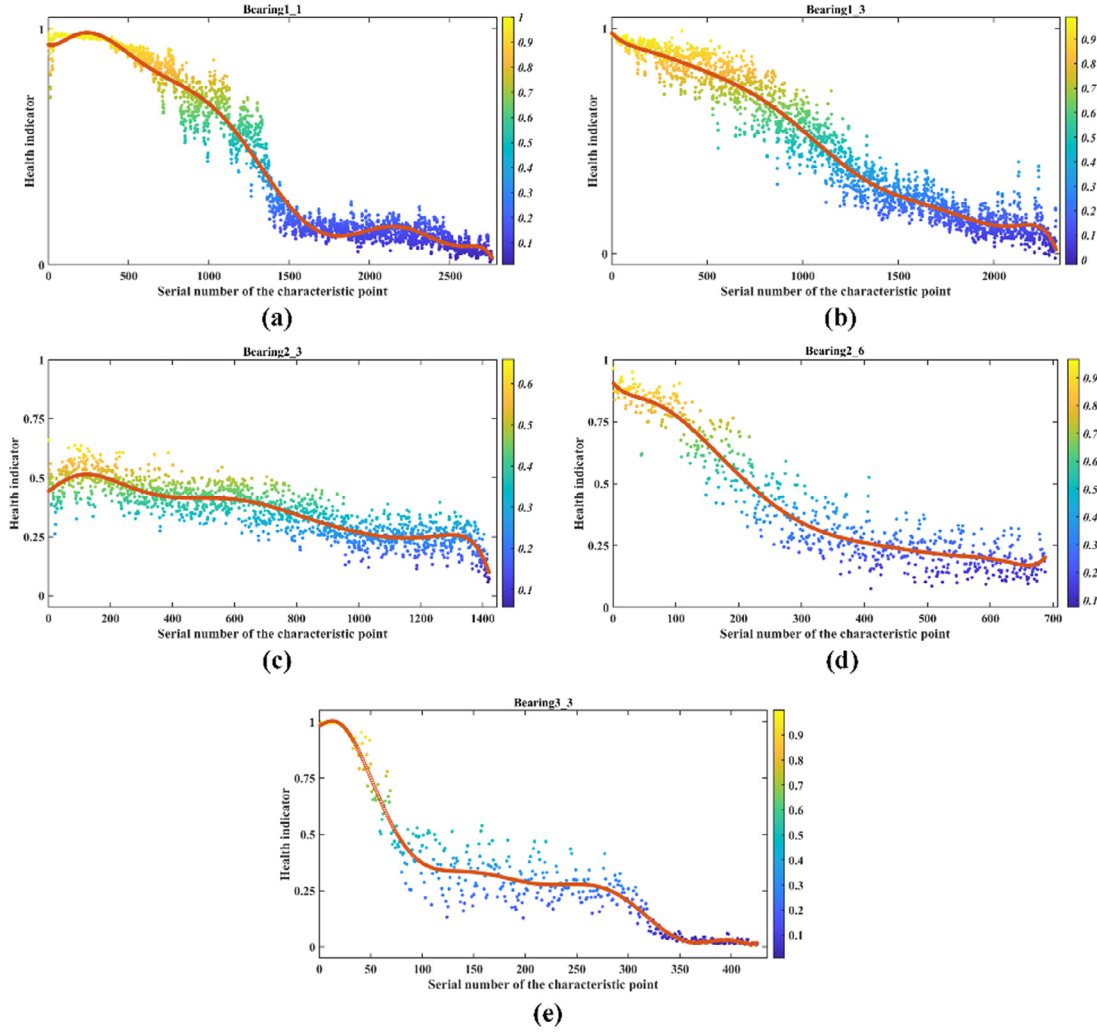


Fig. 9. The HIs of test bearings. (a) bearing1_1. (b) bearing1_3. (c) bearing2_3. (d) bearing2_6. (e) bearing3_3.

(GAU) [11], is able to simultaneously pay attention to the output of the reset and update gates via an attention mechanism, which guarantees the differentiated learning of the previous and current information. Furthermore, the GAU network has been successfully utilized to predict RUL of bearing in Ref. [11]. Thus, the GAU network is selected as the prediction model in this work to show that the HI vector constructed by the quadratic function-based DCAE neural network is suitable for the bearing RUL prediction. The framework of GAU is shown in Fig. 13, and its details can be seen in Ref. [11].

The HIs of bearing1_1 and bearing1_3 constructed by the proposed DCAE neural networks are fed into the GAU model for RUL prediction. A standard initialization method is used to initialize the parameters of the GAU neural network and the failure threshold is taken as 0. Then, a gradient descent algorithm is chose to update the parameters. The flowchart of RUL prediction used in this work is shown in Fig. 14, and bearing1_1 dataset and bearing1_3 dataset are selected for experiment. In bearing1_1 dataset, the first 2611 known HI points of the HI is selected to form training set for predicting 150 unknown HI points. In bearing1_3 dataset, the first 2175 known HI points are applied to train GAU and the next 150 unknown HI points are predicted by the trained GAU. The parameters of GAU neural network are listed in Table 2.

Fig. 15(a) shows the experimental result of bearing1_1, while Fig. 15(b) shows the experimental result of bearing1_3. It can be

Table 2

Main parameters of GAU neural network.

Parameters	Bearing1_1	Bearing1_3
The number of neurons in input layer	160	163
The number of neurons in hidden layer	22	22
The number of neurons in output layer	1	1
Learning rate	0.06	0.06

easily noted from Fig. 15 that the error between the training and the actual HI value is much smaller, and the predictive HI value is close to the actual HI value. Furthermore, the variation trends of both the training and predictive HI curves conform to that of the actual HI curve. It also indicates that the HI vector constructed by quadratic function-based DCAE neural networks can be effectively applied to the bearing RUL prediction. To explain the necessity of using GAU, a widely-used neural network-DNN and the polynomial segment fitting method are also used for the HI prediction. The experimental results obtained by the DNN network and the polynomial segment fitting method are respectively shown in Figs. 16 and 17. From these two figures, we can see that the error between the predicted HI value and the actual HI value is greatly large, and the trend of predicted HI curve is not accord with that of the actual HI curve. Especially, the predicted HI value obtained by the DNN network cannot converge to the failure threshold. Although the predicted HI value

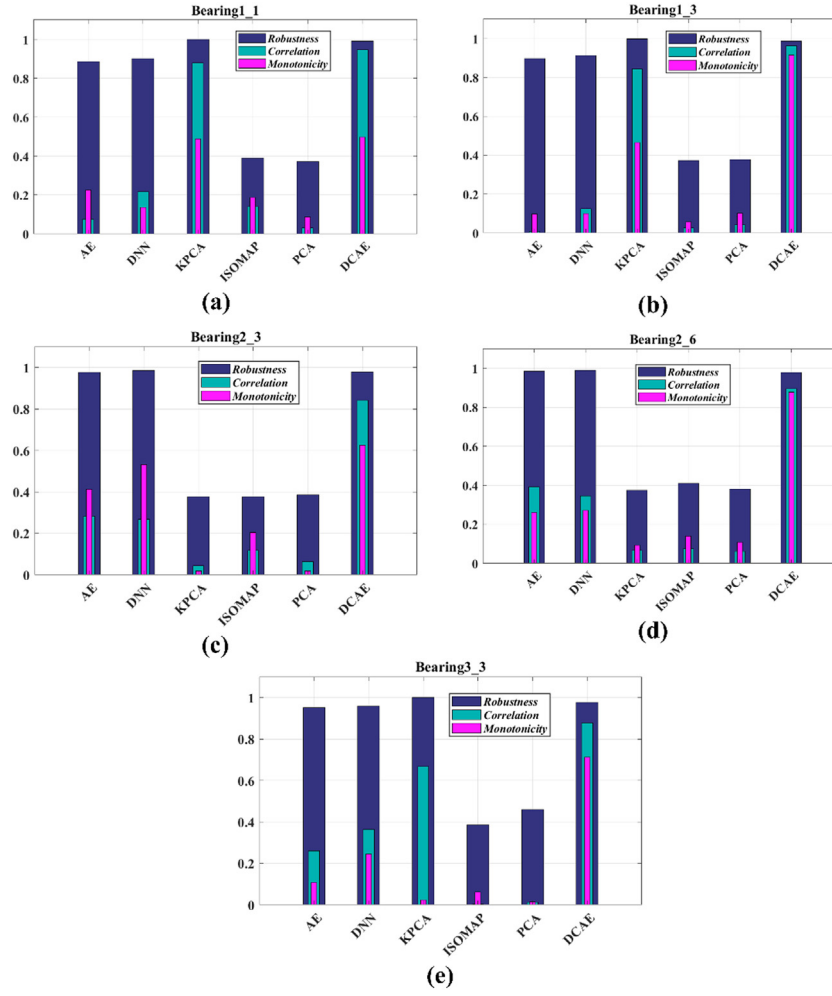


Fig. 10. Three assessment indexes of different methods: (a) bearing1_1. (b) bearing1_3. (c) bearing2_3. (d) bearing2_6. (e) bearing3_3.

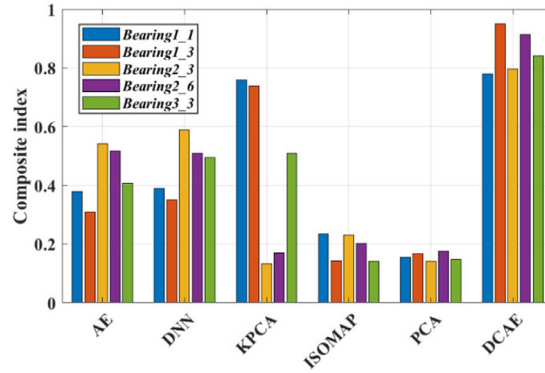


Fig. 11. The composite index of different methods.

obtained by the polynomial segment fitting method can reach the failure threshold, the predicted RUL is much smaller than the actual RUL. This is mainly because that the HI values have a large fluctuation. Consequently, the GAU network is more suitable for the time series prediction than the DNN network and the polynomial segment fitting method.

Next, for showing the superiority of the quadratic function-based DCAE neural network, the constructed HI of bearing1_1 based on AE, DNN and KPCA are used for comparison. The constructed HIs based on ISOMAP and PCA are not utilized for RUL

prediction because the HIs constructed by ISOMAP and PCA cannot represent the degradation trend of bearing1_1, as shown in Fig. 18. Five assessment indexes including Score function, mean absolute error (MAE), normalized root mean square error (NRMSE), root mean square error (RMSE) and mean absolute percentage error (MAPE), are utilized to comprehensively assess the predictive ability. The score function, MAE, MAPE, NRMSE and RMSE are defined in Ref. [11]. To perform a fair comparison, we separately optimize the parameters of GAU neural networks

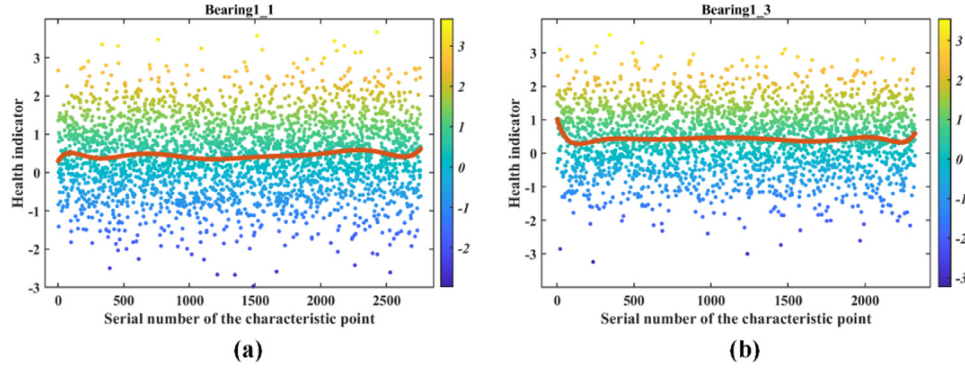


Fig. 12. The constructed HIs of bearing1_1 and bearing1_3 based on VAE. (a) bearing1_1. (b) bearing1_3.

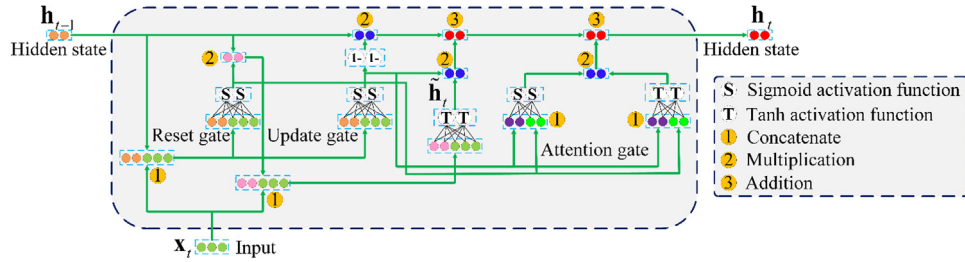


Fig. 13. Gated Attention Unit (GAU).

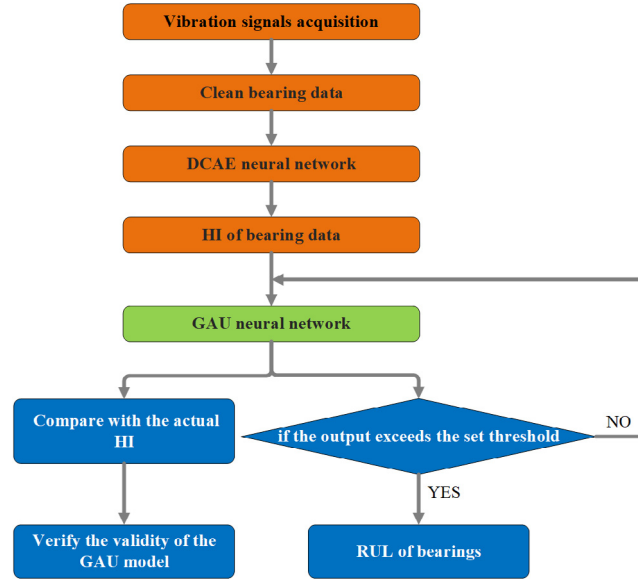


Fig. 14. The flowchart of RUL prediction method for rolling bearings.

according to different HI vectors, and we perform 5 times predictions on each HI vector for obtaining a result of high confidence. The HIs constructed by four different methods are applied to RUL prediction, and the obtained results are illustrated in Fig. 19. We can note from this figure that the DCAE-based HI (DCAE-HI) has the maximal score and minimum prediction error compared to the traditional approaches. Especially, DCAE-HI is much better than AE-based HI (AE-HI), DNN-based HI (DNN-HI) and KPCA-based HI (KPCA-HI) in terms of MAE, NRMSE, RMSE and MAPE. This indicates that the proposed HI based on the quadratic function-based DCAE neural network is more suitable for predicting RUL and has better prediction performance than other HIs.

6. Conclusions

Most of the HI construction approaches need first extract feature, and then construct HI by some dimensionality reduction algorithms or neural networks. In order to directly construct a good HI from vibration signals, this paper proposes a new quadratic function-based DCAE neural network. Via labeling the output of encoder with a quadratic function, a new loss term is introduced into the loss function of DCAE. Owing to the improved loss function, the monotonicity of the constructed HI can be ensured and enhanced. The comparative results of several bearing datasets verify the superiority of the novel HI over other HIs obtained by AE, deep neural network (DNN), KPCA, and VAE. In particular, *Mon*, *Corr* and *CI* of the proposed HI greatly exceeds

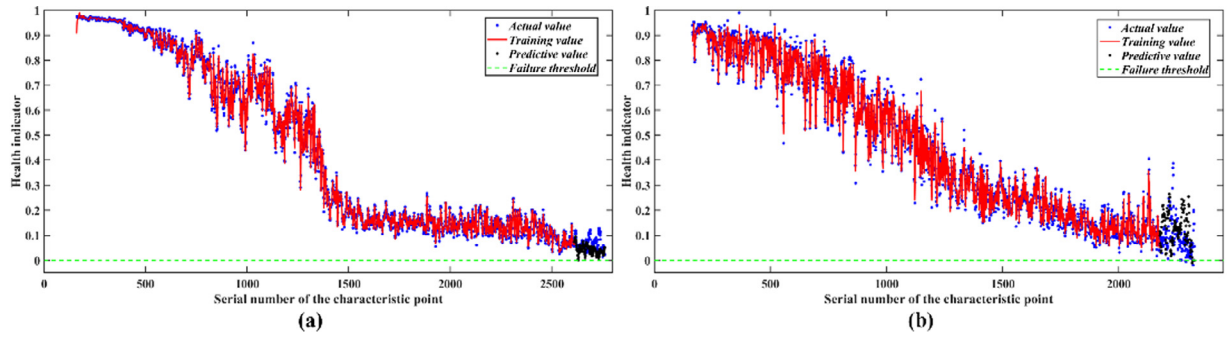


Fig. 15. The experimental results of the GAU network. (a) RUL prediction of bearing1_1. (b) RUL prediction of bearing1_3.

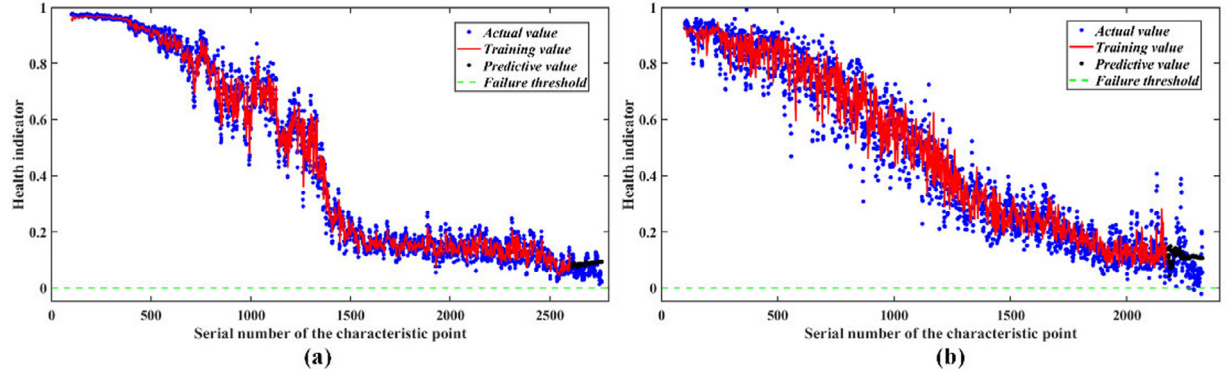


Fig. 16. The experimental results of the DNN network. (a) RUL prediction of bearing1_1. (b) RUL prediction of bearing1_3.

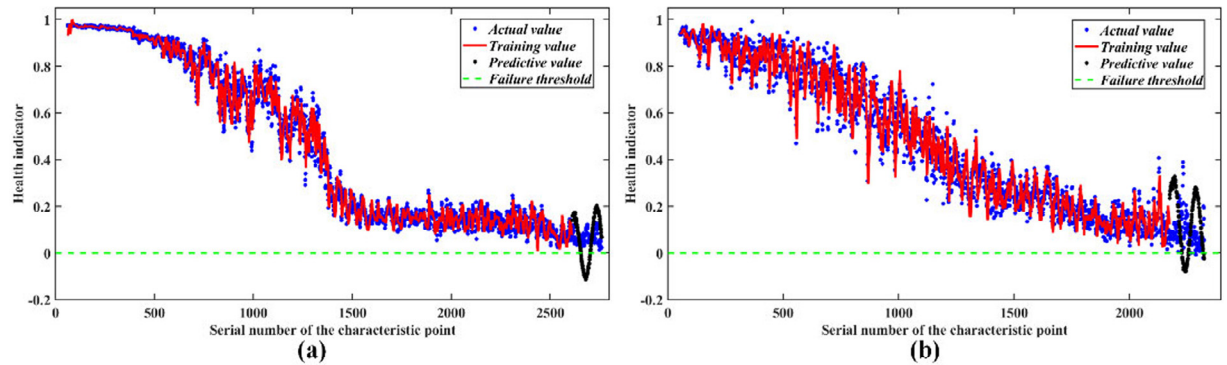


Fig. 17. The experimental results of the polynomial segment fitting method. (a) RUL prediction of bearing1_1. (b) RUL prediction of bearing1_3.

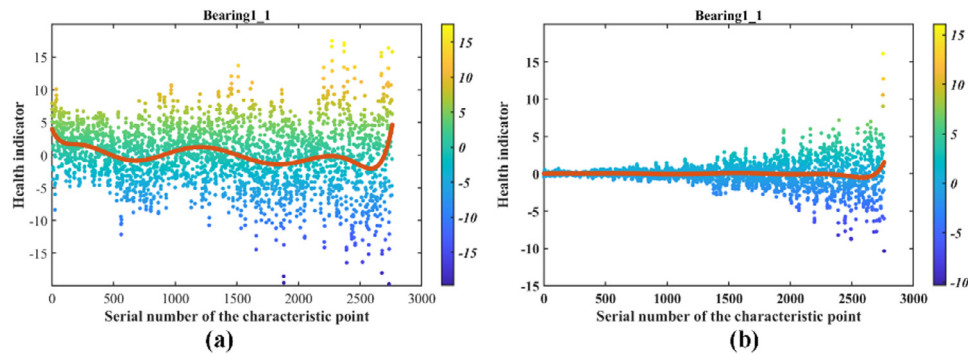


Fig. 18. The constructed HIs of bearing1_1 based on ISOMAP and PCA. (a) ISOMAP. (b) PCA.

other traditional HIs. Moreover, using the proposed HI construction approach, the failure threshold of bearings with general applicability can be easily determined, which can be taken as zero

in most instances. Combining the constructed HI with GAU neural network, the RULs of rolling bearings can be well predicted. The experimental results reveal that the newly developed HI-based

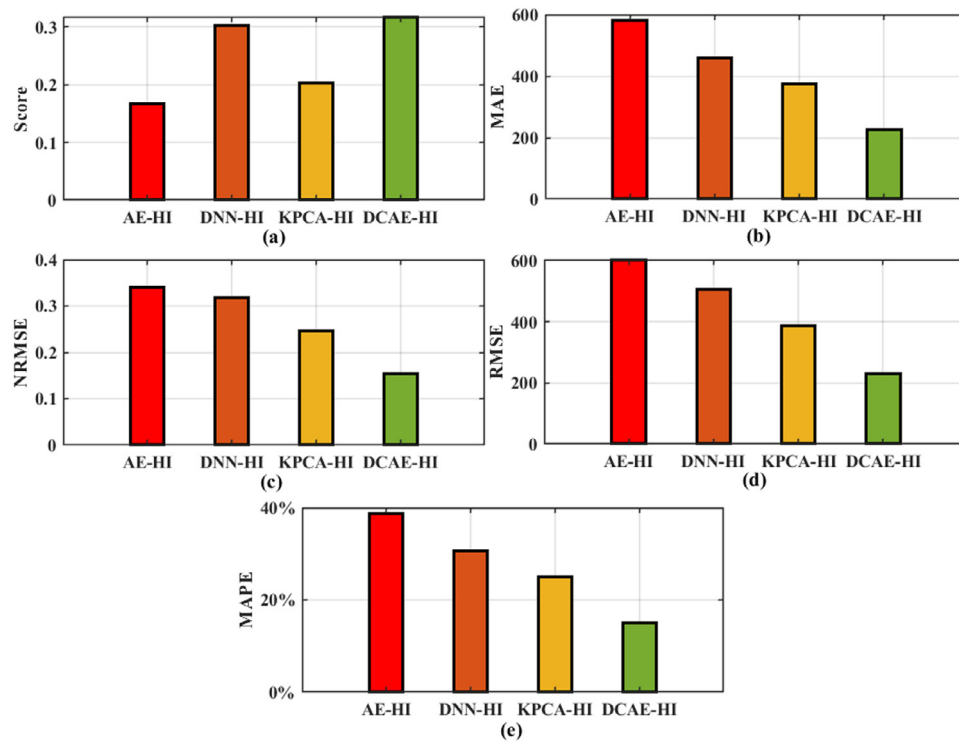


Fig. 19. Evaluation indexes of different HI construction methods.

predictive method has stronger predictive ability than other HIs-based methods. Consequently, this work is of great engineering value in bearing RUL prediction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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